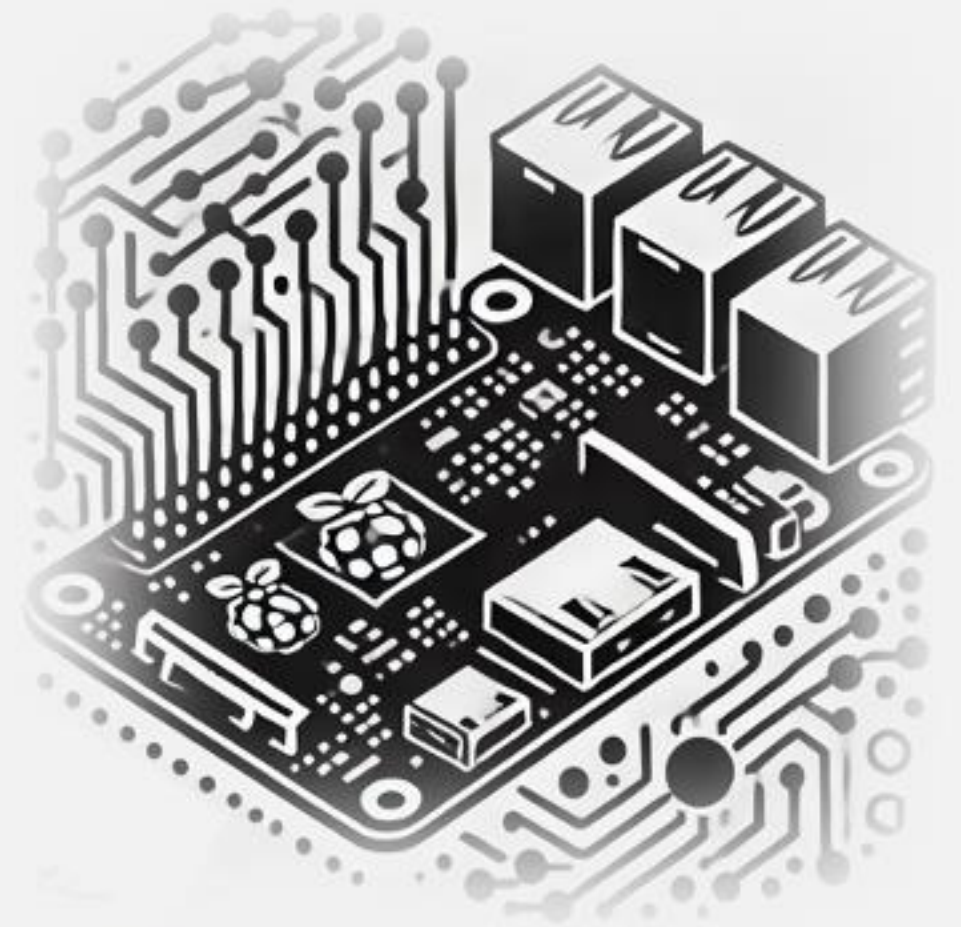


IESTI05 – Edge AI

Machine Learning System Engineering

-
1. Introduction to Edge AI
About the Course & Syllabus



Who I am

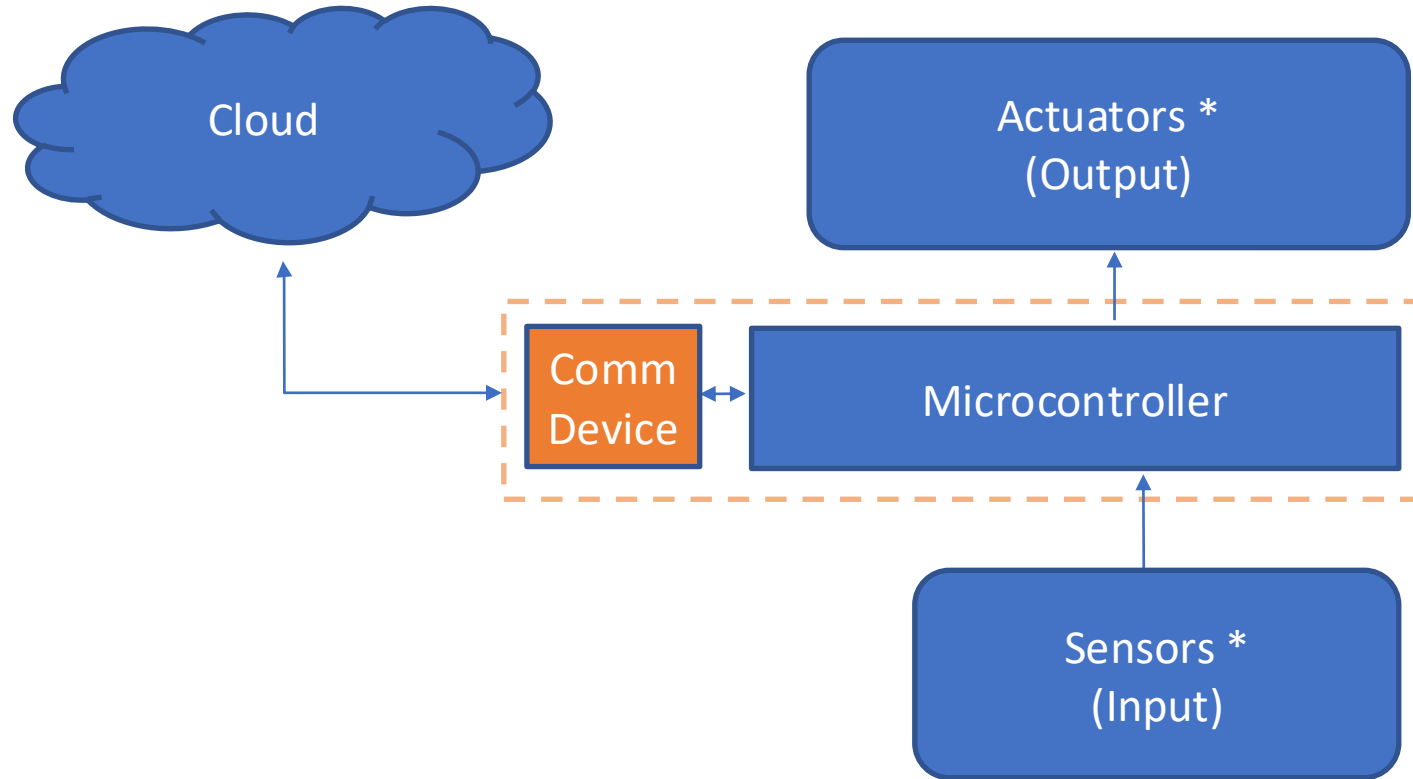
- Brazilian from São Paulo, **Data Science Master's degree from UDD**, Chile, and **MBA from IBMEC (INSPER)**, Brazil.
- Graduated in 1982 as an **Engineer from UNIFEI** with **Specialization from Poli/USP**, both in Brazil.
- Worked as a **teacher, engineer, and executive** in several technology companies such as CDT/ETEP, AVIBRAS Aeroespacial, SID Informática, ATT-GIS, NCR, DELL, COMPAQ (HP), and more recently at IGT as a Regional VP, and a Senior Advisor for Latin America.
- **Write about electronics**, publishing in sites as MJRoBot.org (Editor/Writer), Hackster.io (#1 Contributor), Instructables.com, and Medium.com (TDS – Towards Data Science).
- **Professor** at UNIFEI (IESTI), teaching “Machine Learning applied to Embedded Devices” course (IESTI01) and EdgeAI Engineering (IESTI05).
- **AiEng4D group** and the **EDGE AIP** – the Academia-Industry Partnership of **EDGEAI Foundation** Co-Chair, promoting EdgeAI education in developing countries.



Marcelo Rovai

Internet of Things (IoT)

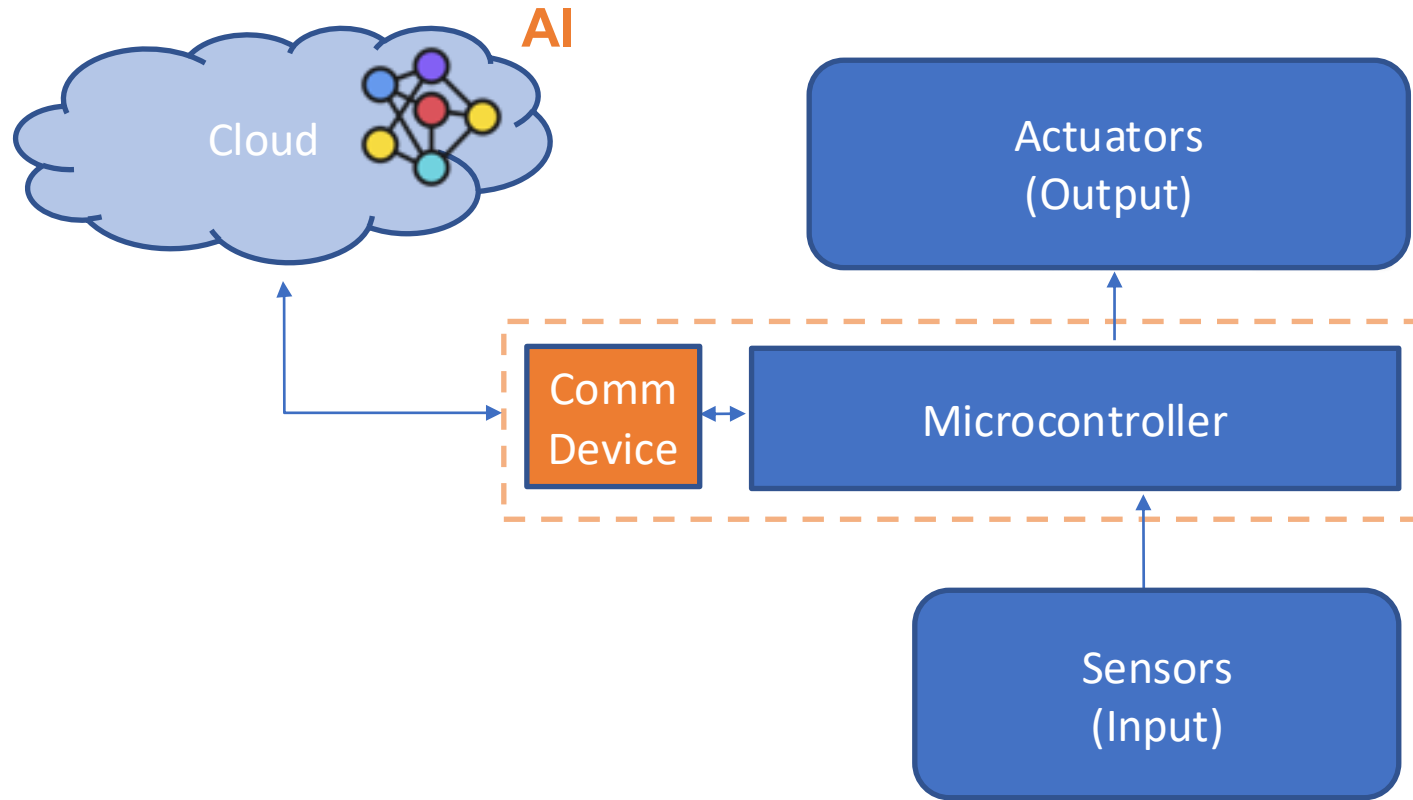
Typical IoT Project



* "Things"



Typical **A**IoT Project ...



... **Issues**

Bandwidth

Latency

Energy

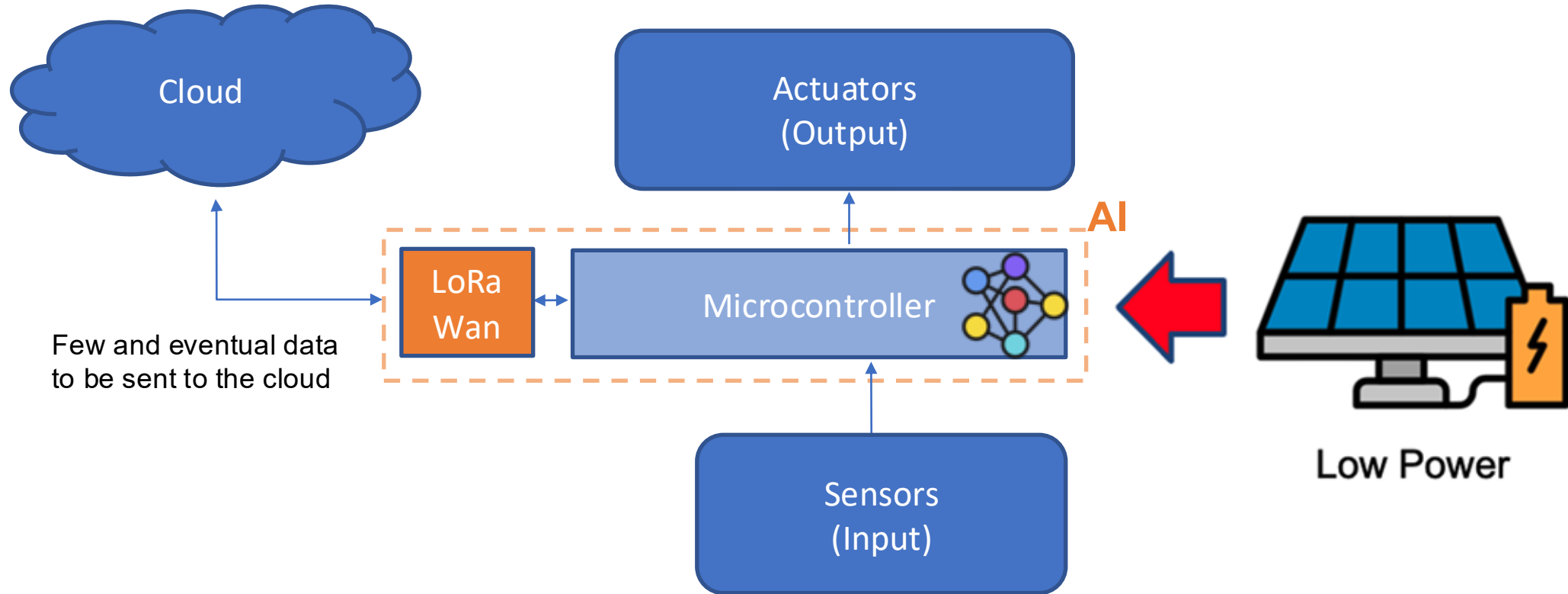
Reliability

Privacy

... **S**olution ?

IoT 2.0 * – Edge AI/ML

* Intelligence of Things

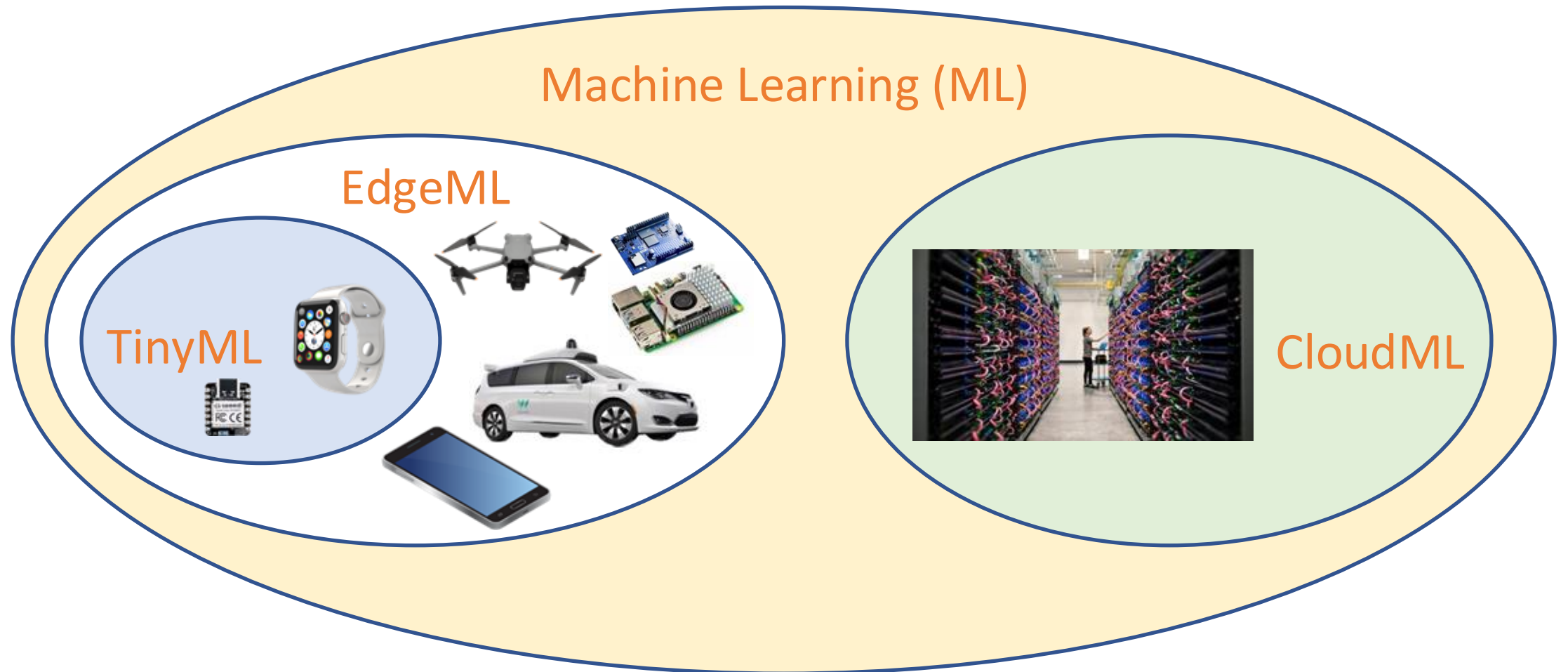


... **Solution** -> ML goes close to data

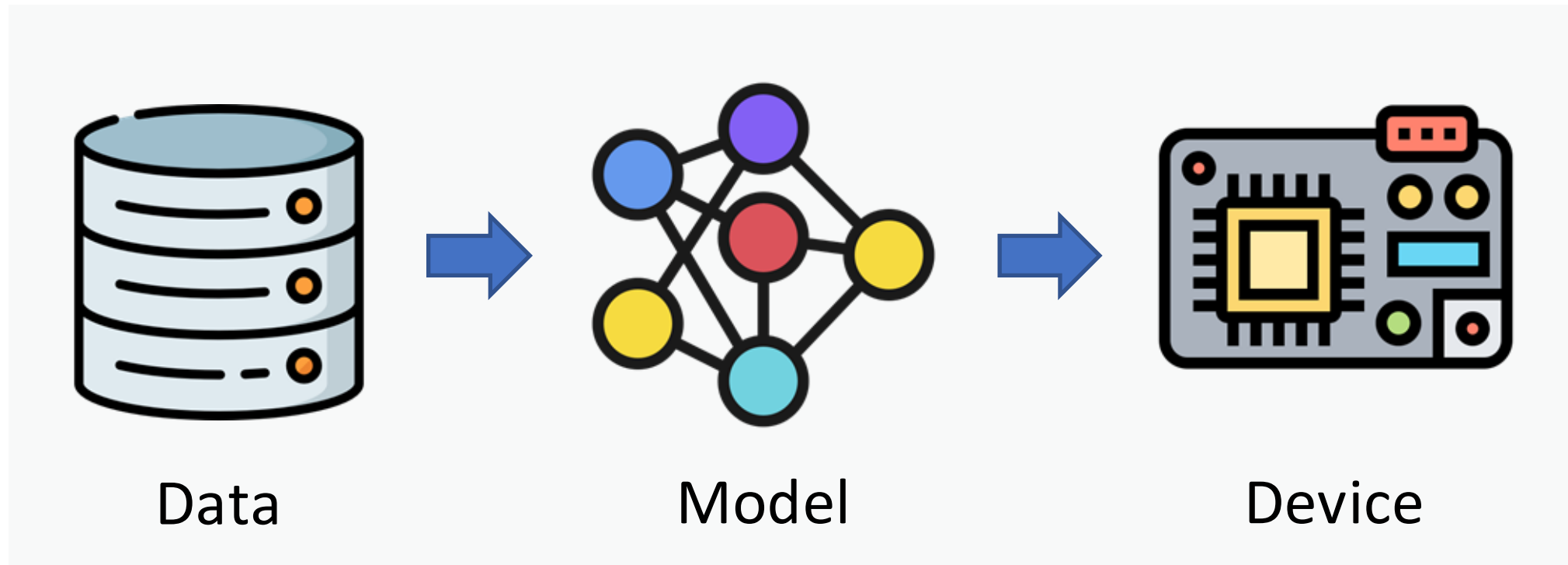
Embedded ML

EdgeAI & TinyML

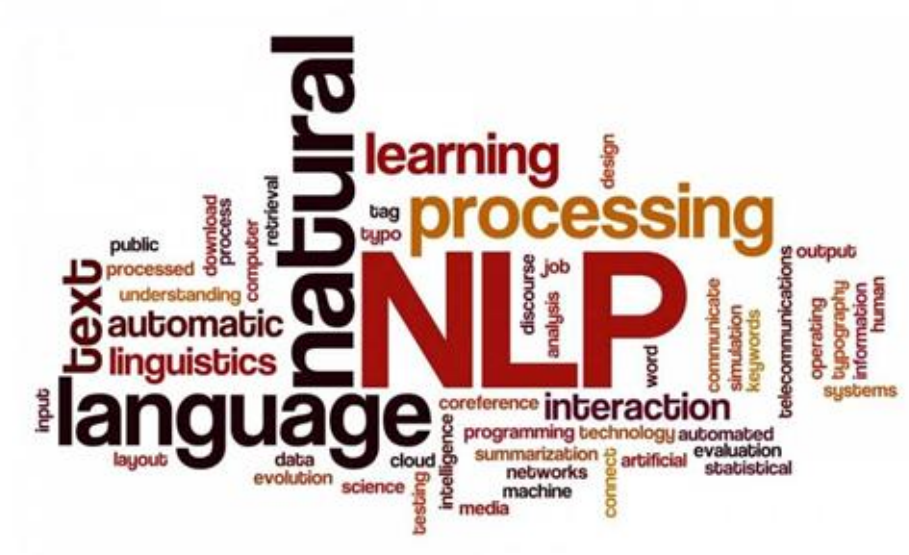
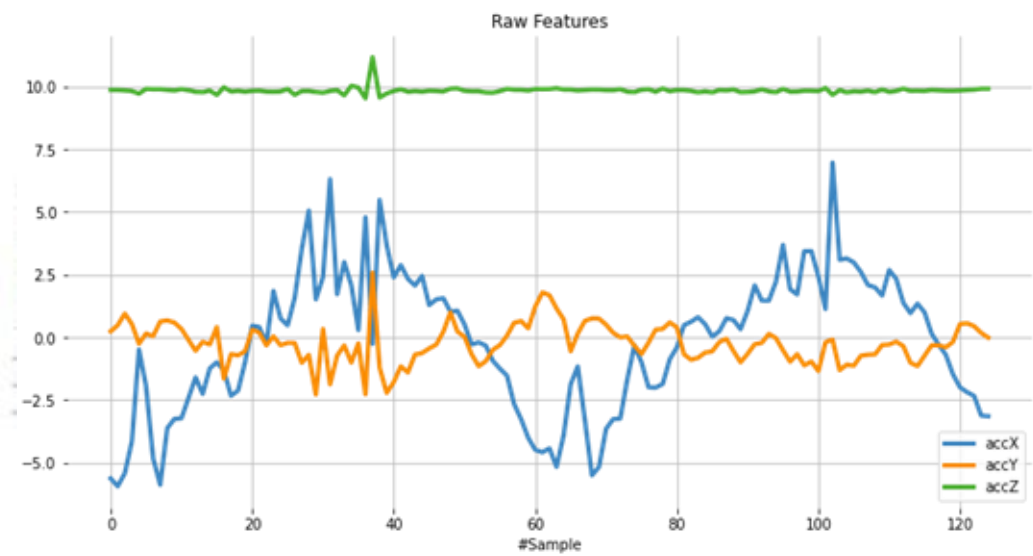
From Tiny to Cloud: One AI, Many Scales



ML Deployment Pipeline



Unstructured Data





Vibration
Analysis

Image
Classification

Text
Generation

Image
Generation

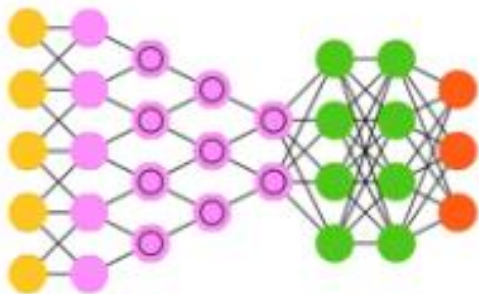
Large Language
Models- LLMs



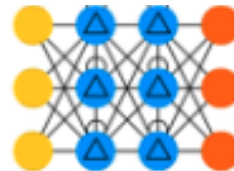
MLP - Deep Neural Network



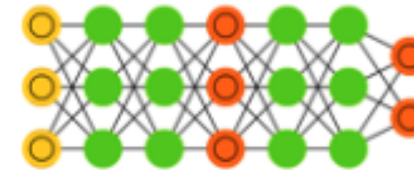
CNN - Convolutional NN



RNN - Recurrent NN (GRU/LSTM)



GAN - Generative Adversarial Net.



TF - Transformer (Attention)



Datasets Preprocessing

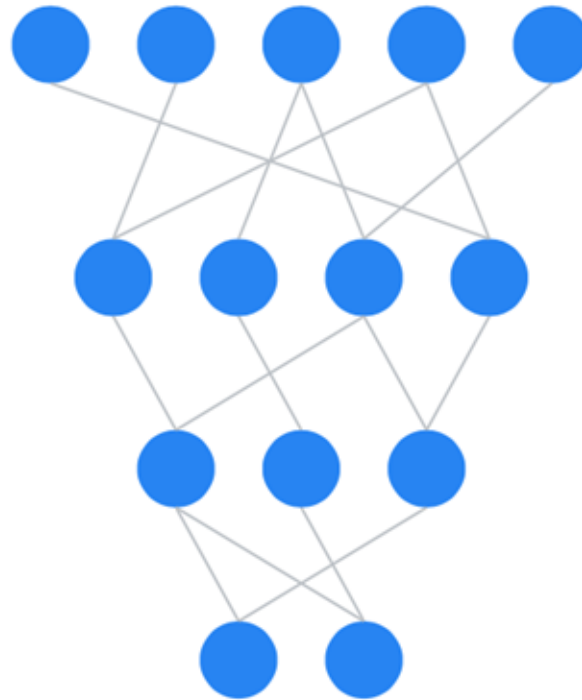
Sound

Vision

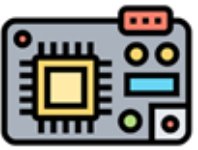
Vibration

Text

Quantization Pruning, Distillation

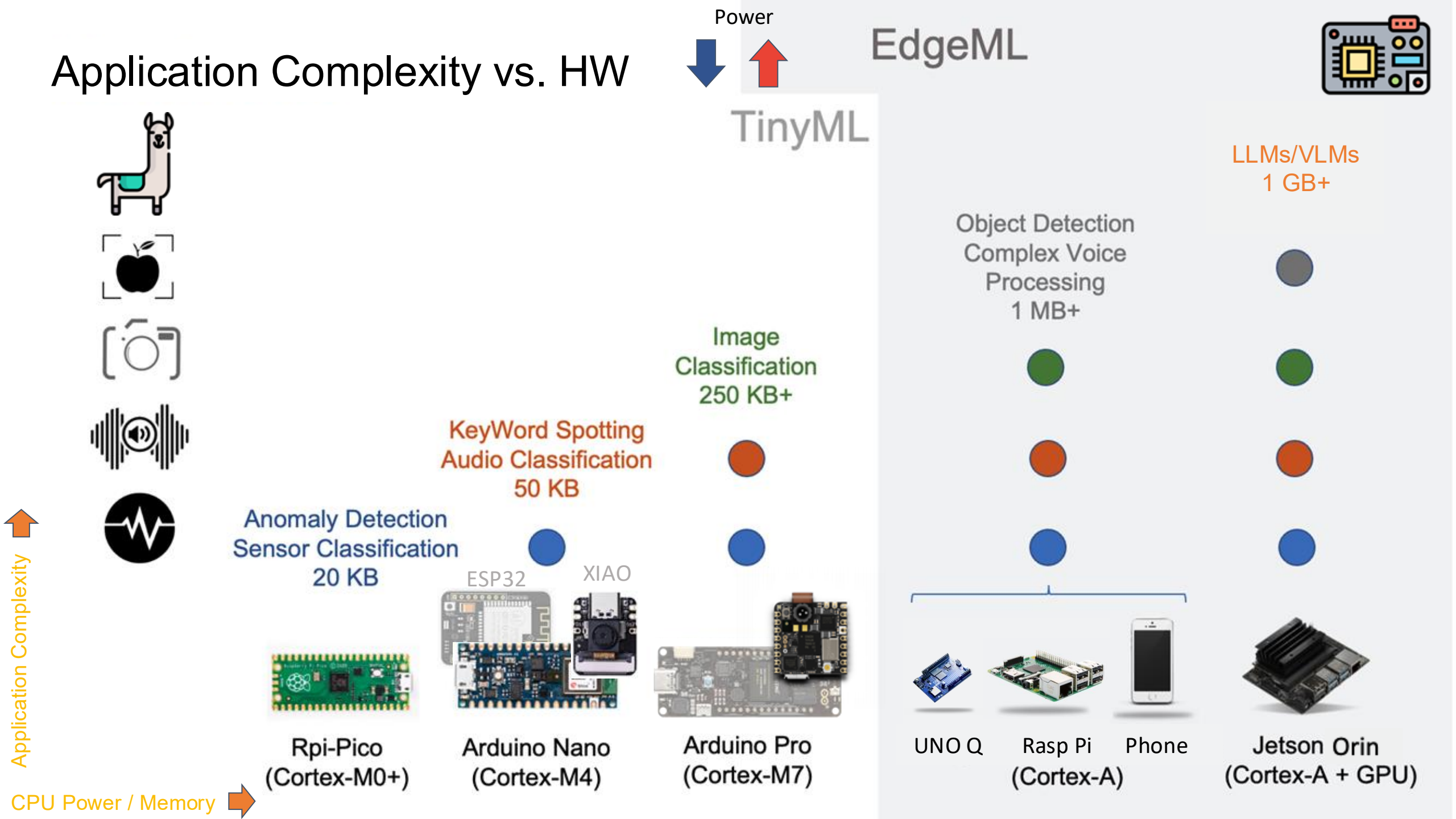


Resource constraints

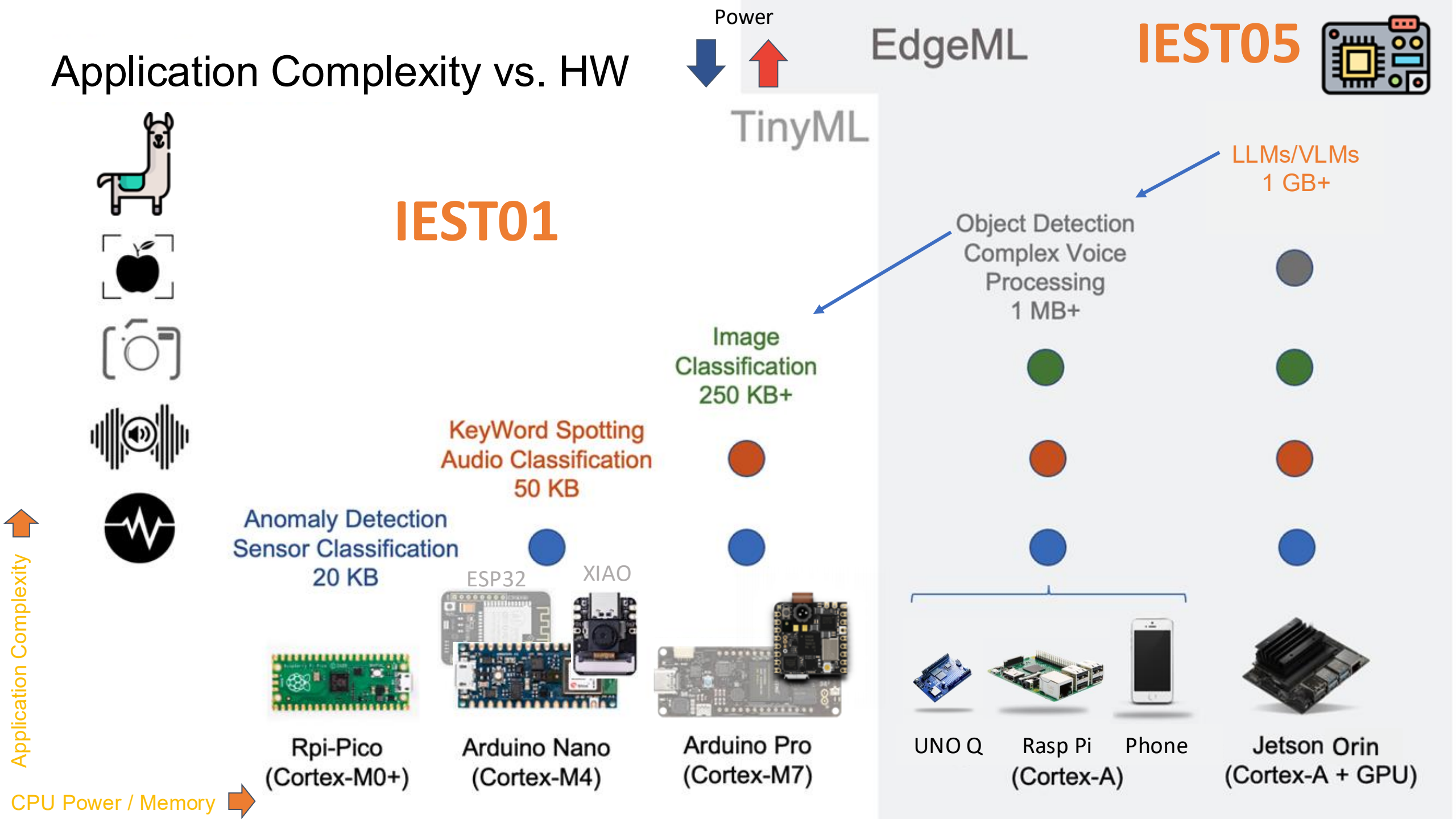


End-to-end TinyML/EdgeAI Application

Application Complexity vs. HW



Application Complexity vs. HW



Generative AI at the Edge

Generative AI (GenAI)

Generative AI is an artificial intelligence system capable of creating new, original content across various media — **text, images, audio, and video**. These systems learn patterns from existing data and use that knowledge to **generate novel outputs that didn't previously exist**.

When used for generative tasks, these model families are all types of GenAI:

LLM

**Large
Language
Models**

GPT, Claude,
Gemini, Qwen

SLM

**Small
Language
Models**

Gemma, Phi,
Llama, Qwen

VLM

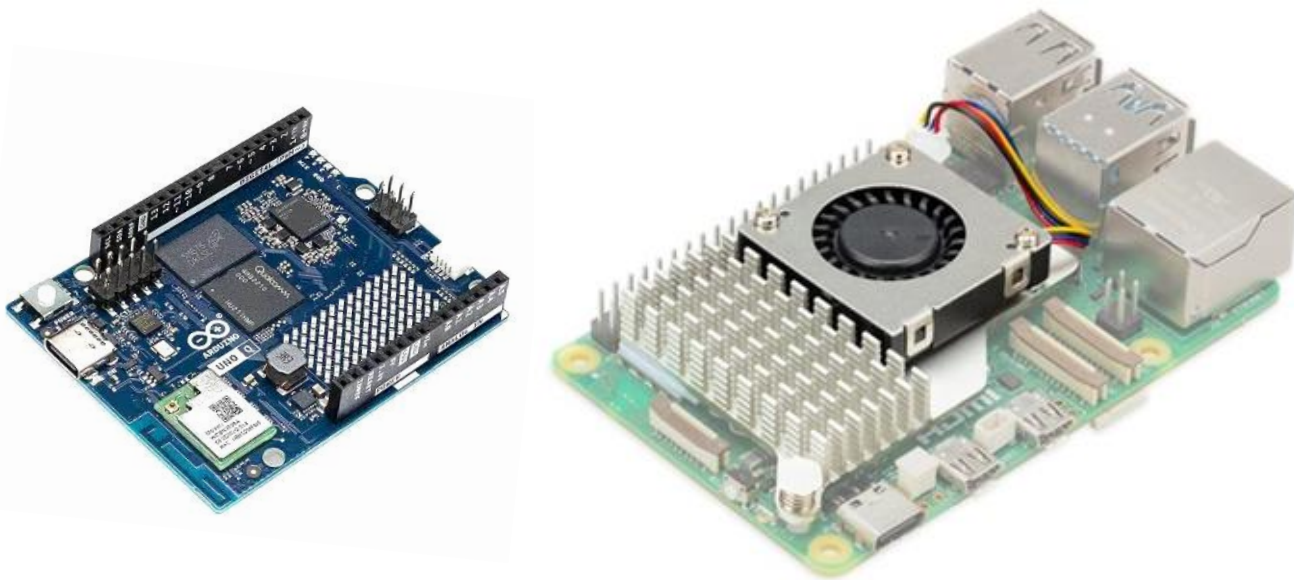
**Visual-
Language
Models**

Gemini, Gemma,
MoonDream, GPT,
PaliGemma,
Florence

GenAI at the Edge

"GenAI at the Edge means the model lives on the device — not in a data center."

While LLMs typically require cloud infrastructure, compact SLMs* and small VLMs can run directly on edge devices.



* Current trends point to 2026 as the year of SLMs (Models < 7B to 10B), with specialized models surpassing generalist ones in specific tasks, such as programming and medical diagnoses.

Sustainability Matters

0.5W



5W-



15W



20W+



100W+



100x more energy efficient than cloud AI



Vibration
Analysis

Image
Classification

Text
Generation

Image
Generation

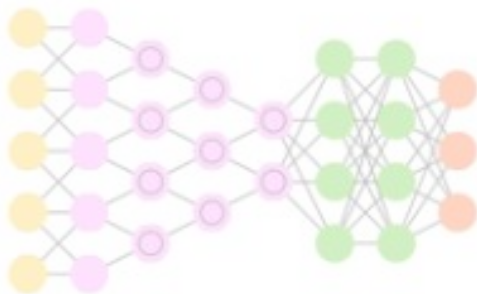
Large Language
Models- LLMs



MLP - Deep Neural Network



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GAN - Generative Adversarial Net.



TF - Transformer (Attention)



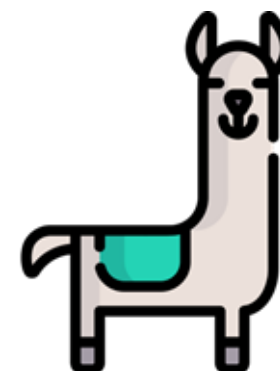
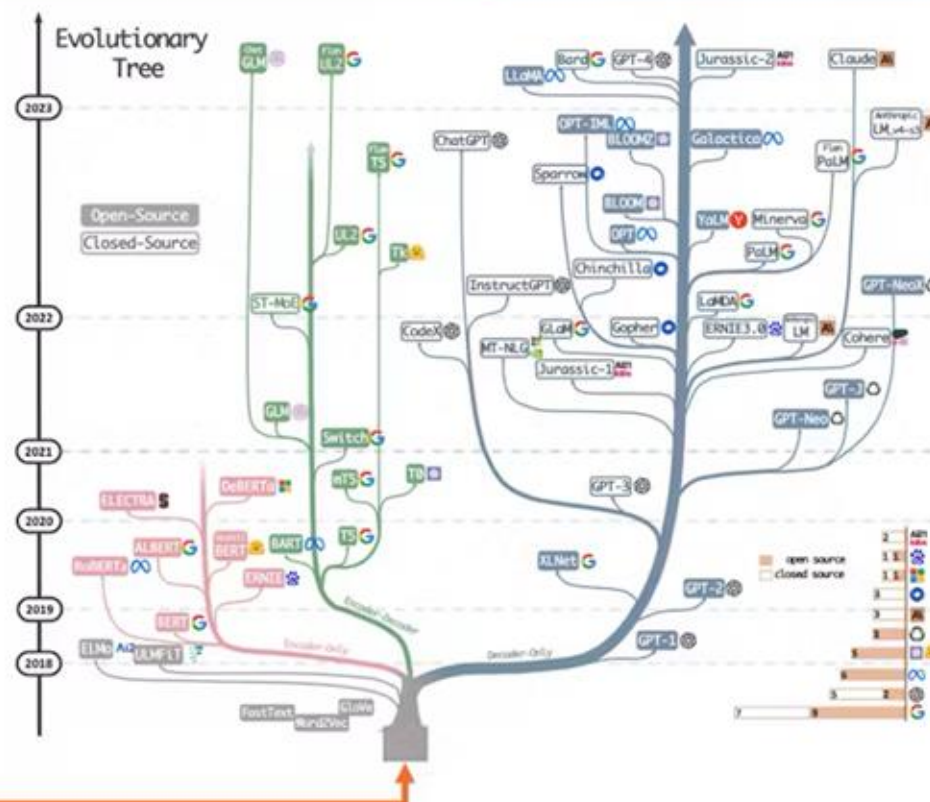
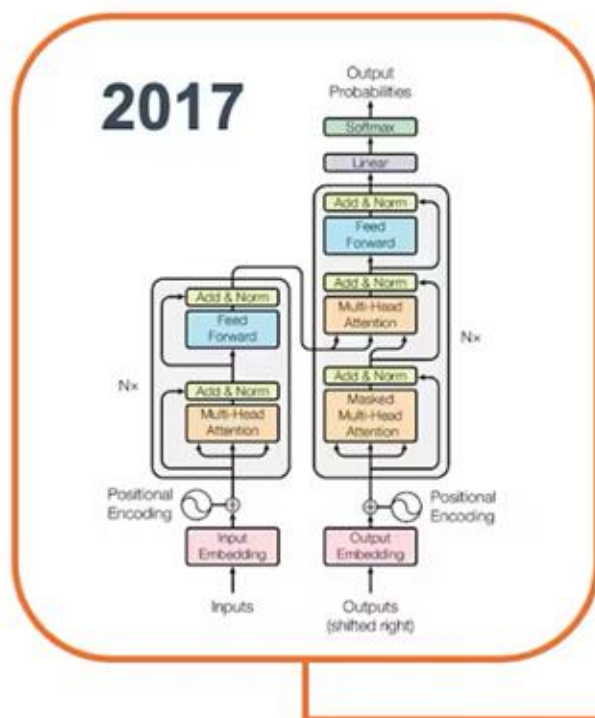
Transformers to LLMs and SLMs

Open - China

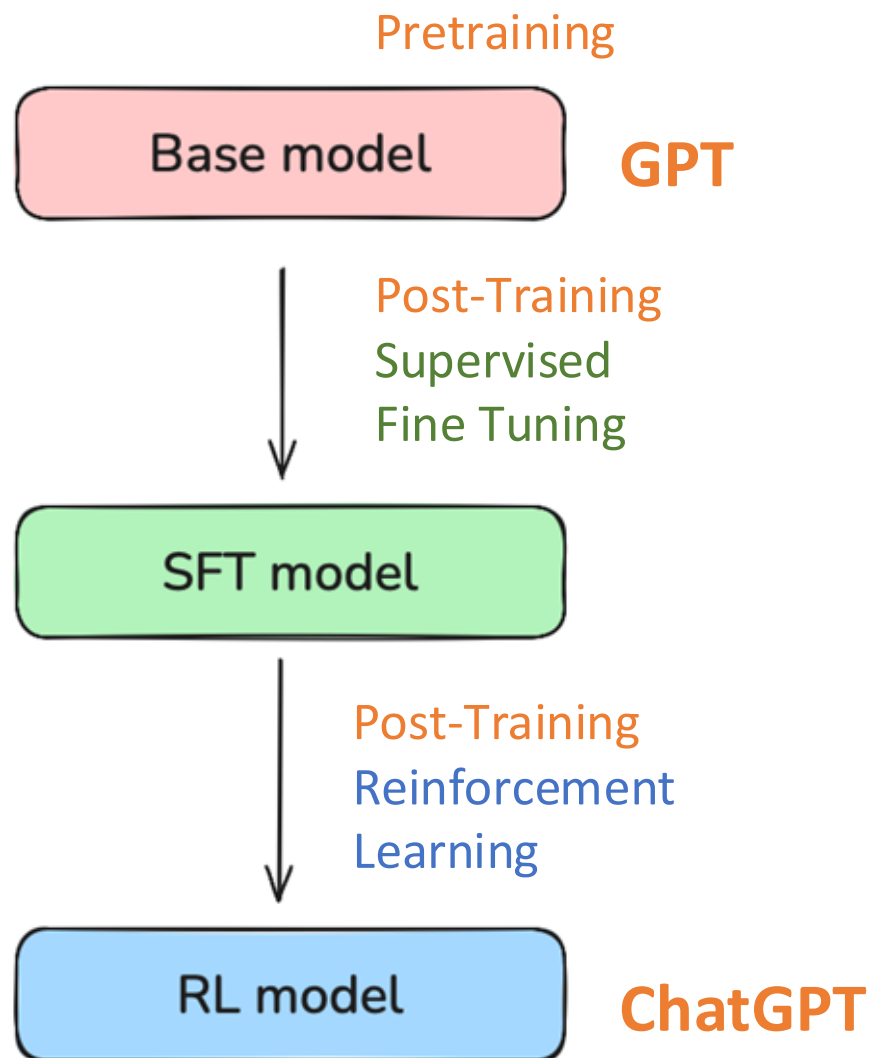
Open

Closed

2026

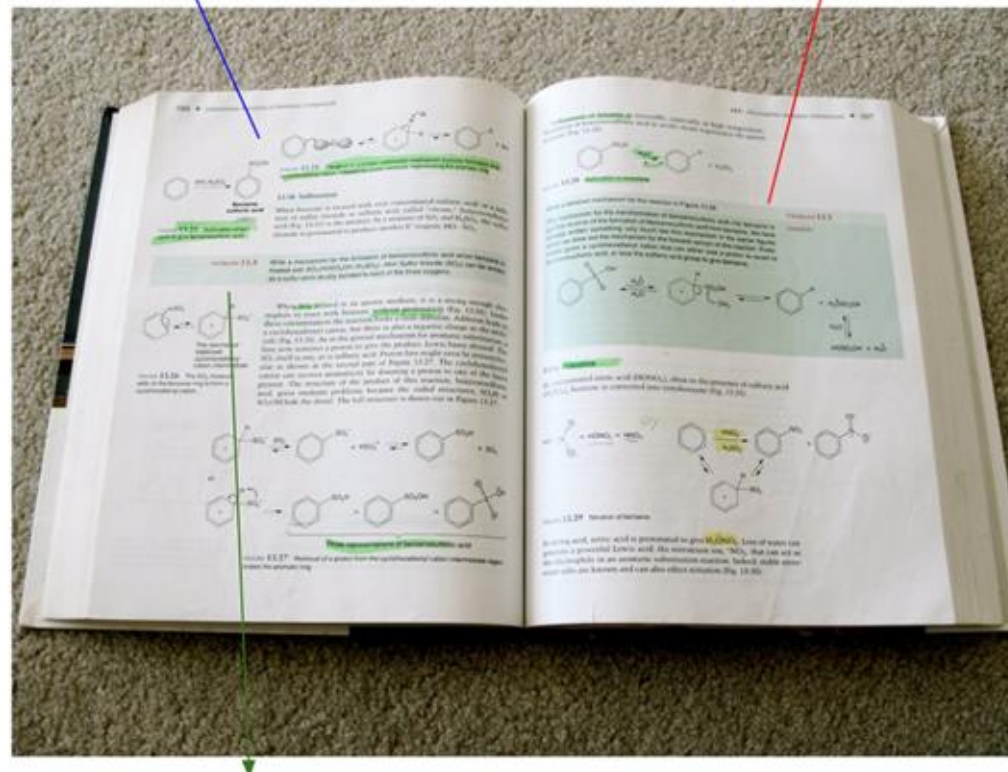


LLM TRAINING



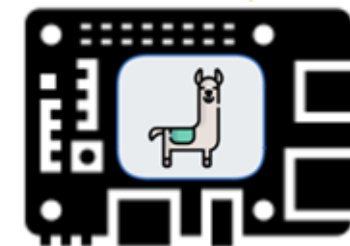
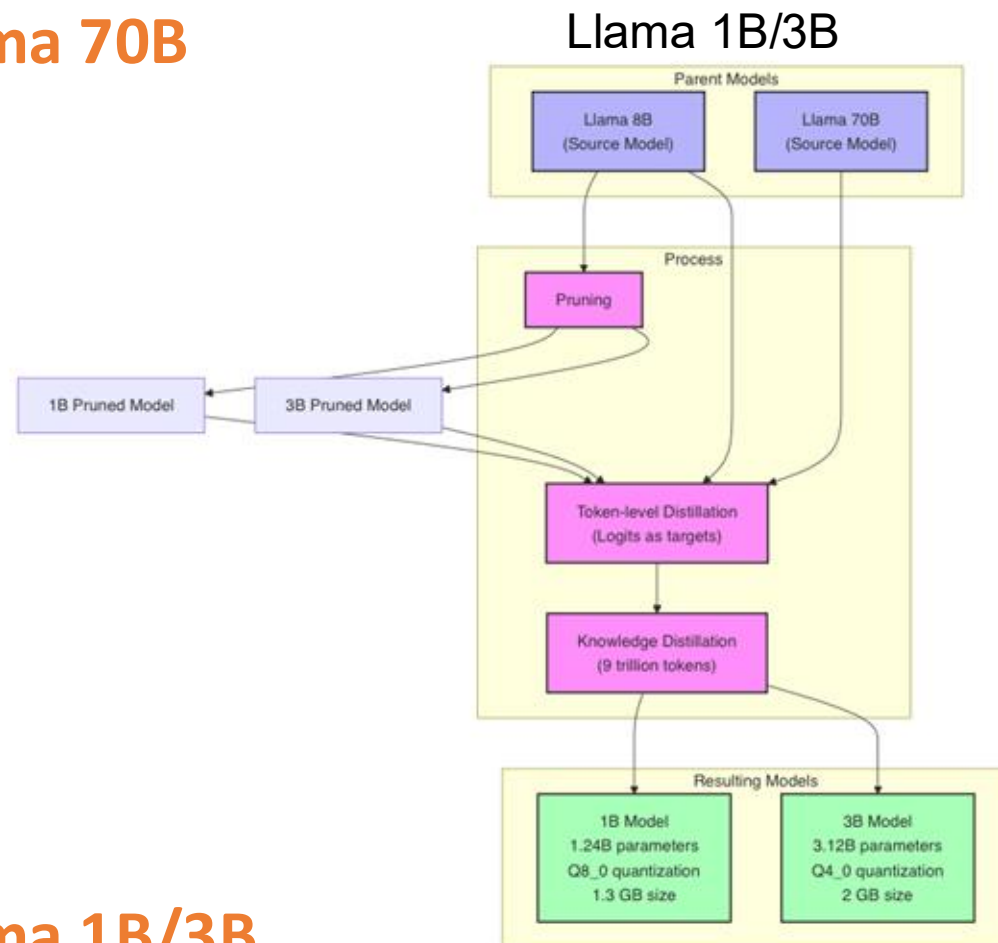
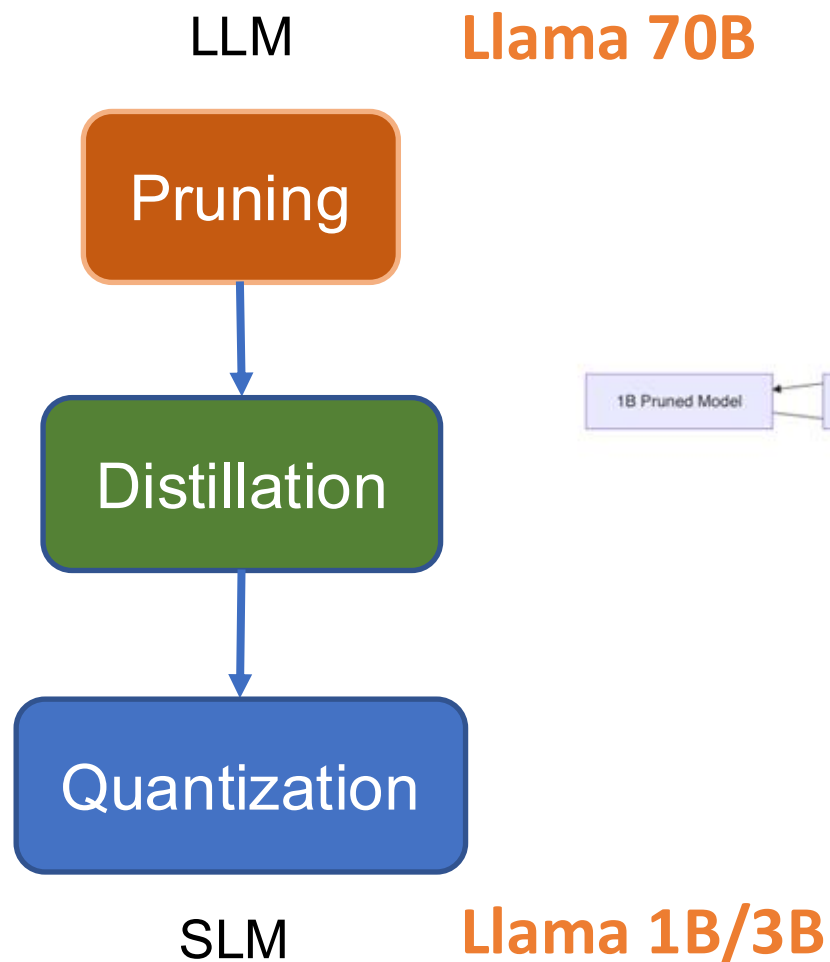
exposition $\Leftrightarrow$ pretraining
(background knowledge)

worked problems $\Leftrightarrow$ supervised finetuning
(problem + demonstrated solution, for imitation)



practice problems $\Leftrightarrow$ reinforcement learning
(prompts to practice, trial & error until you reach the correct answer)

Small Language Models (SLMs)



Running SLMs locally

How to Run SLMs at the Edge

Ollama

Open-source framework for running SLMs locally. The easiest way to get started on Raspberry Pi — supports Qwen, Phi, Gemma, Llama, LLaVa, and more. Does not run easily on Arduino UNO-Q.

llama.cpp

Lightweight C++ inference engine for quantized models. Tested successfully on Arduino UNO-Q — preferred for more control, compact size, and low memory usage. Also efficient on Raspberry Pi.

HF Transformers

Python library with broad model and task coverage via the Hugging Face ecosystem. Well-supported for running SLMs on Raspberry Pi.

LiteRT-LM

Google AI Edge solution designed for production deployments. Optimized for running language models efficiently across edge devices.

Small Models

LlaMa



Gemma



```
marcelo_rovai — mjrovai@raspi-5: ~ — ssh mjrovai@192.168.4.209 — 74x12
(ollama) mjrovai@raspi-5:~ $ ollama run llama3.2:3b --verbose
>>> What is the capital of France?
The capital of France is Paris.

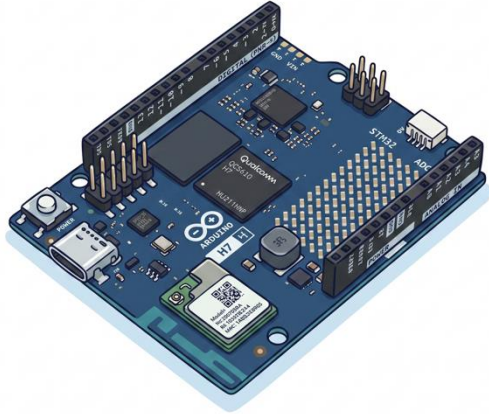
total duration:      1.808927736s
load duration:      39.854862ms
prompt eval count:   32 token(s)
prompt eval duration: 221.506ms
prompt eval rate:    144.47 tokens/s
eval count:          8 token(s)
eval duration:       1.506376s
eval rate:           5.31 tokens/s
```

```
marcelo_rovai — mjrovai@raspi-5: ~ — ssh mjrovai@192.168.4.209 — 67x13
(ollama) mjrovai@raspi-5:~ $ ollama run gemma2:2b --verbose
>>> What is the capital of France?
The capital of France is **Paris**. 🚦

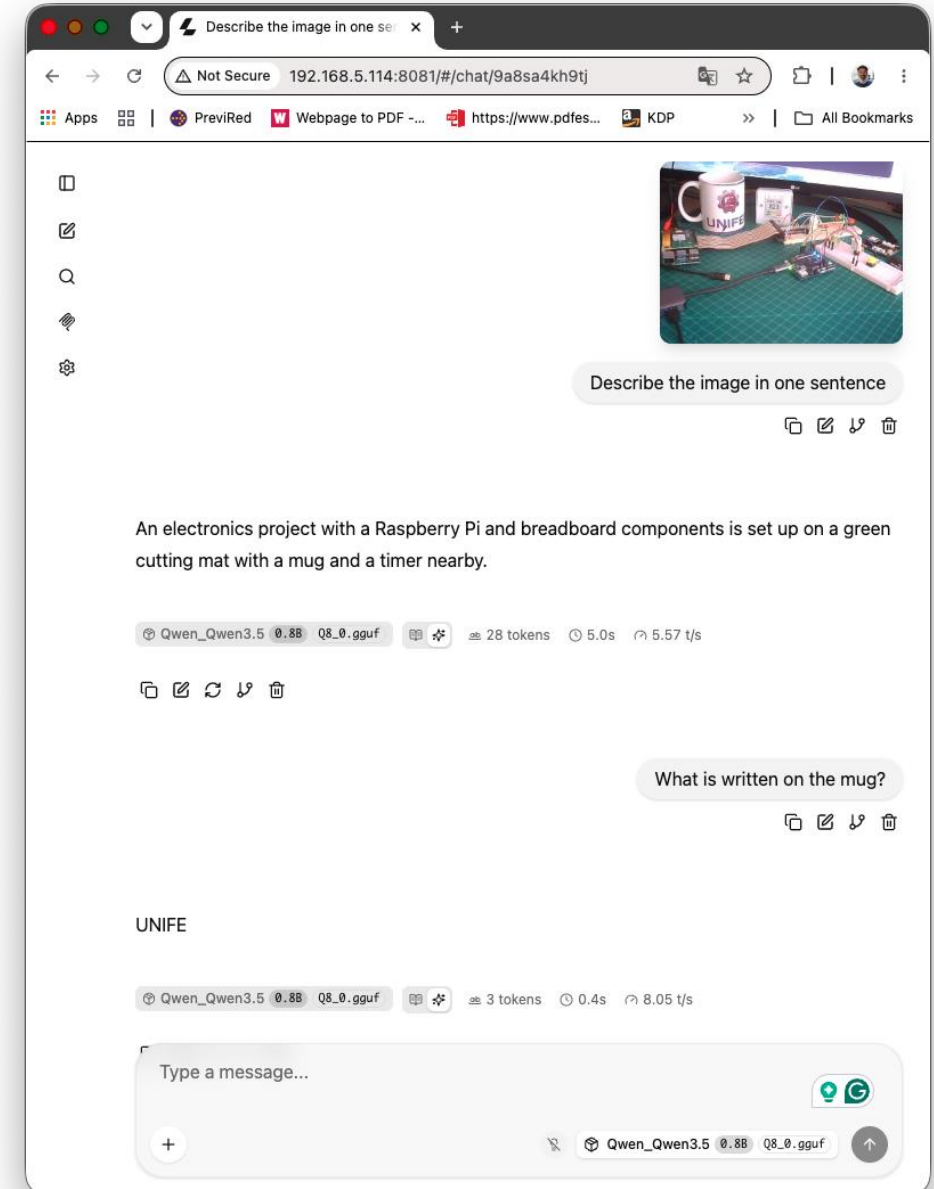
total duration:      4.373339337s
load duration:       48.129697ms
prompt eval count:   16 token(s)
prompt eval duration: 1.968114s
prompt eval rate:    8.13 tokens/s
eval count:          13 token(s)
eval duration:       2.313284s
eval rate:           5.62 tokens/s
```

Small Models

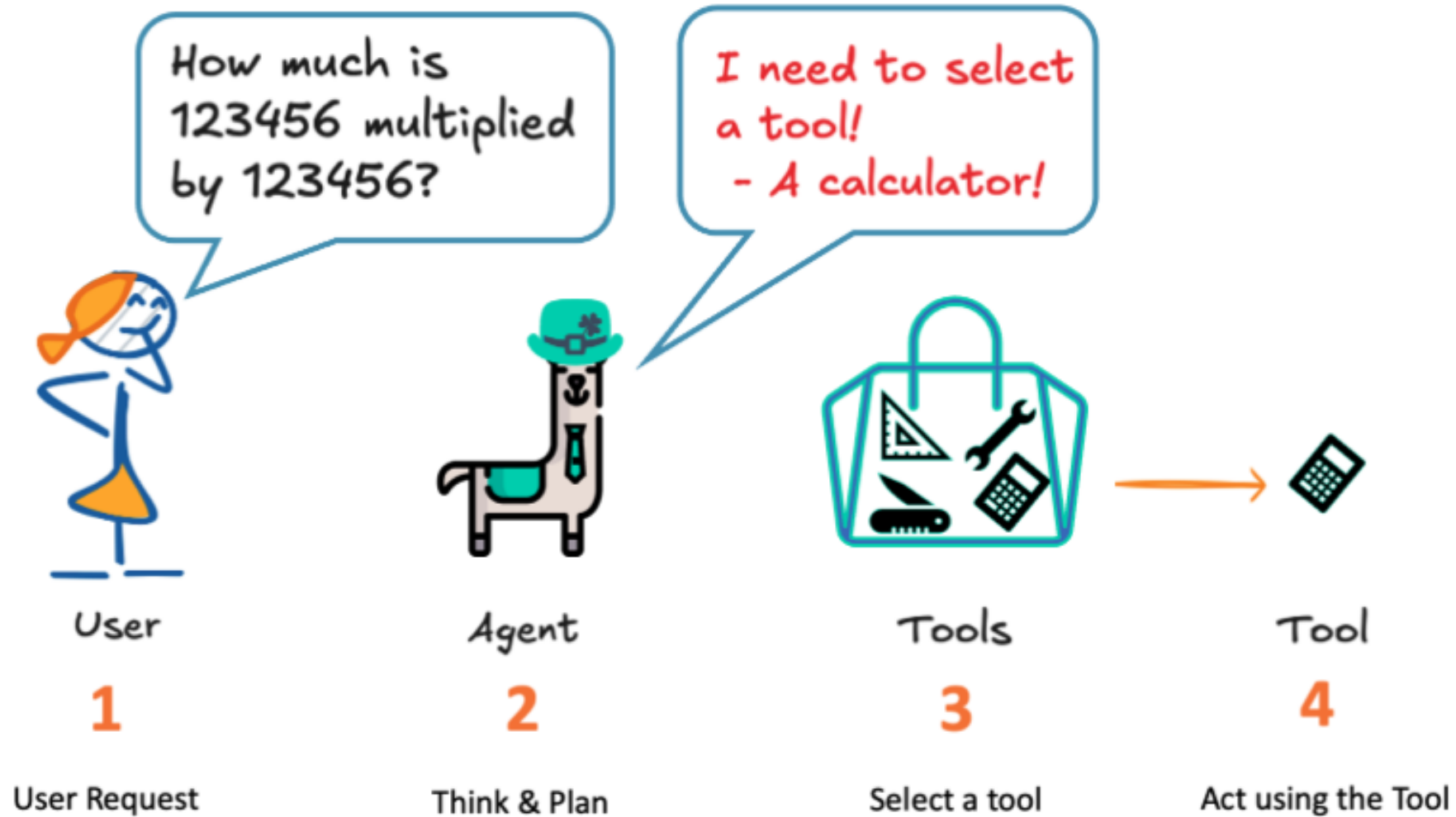
Qwen



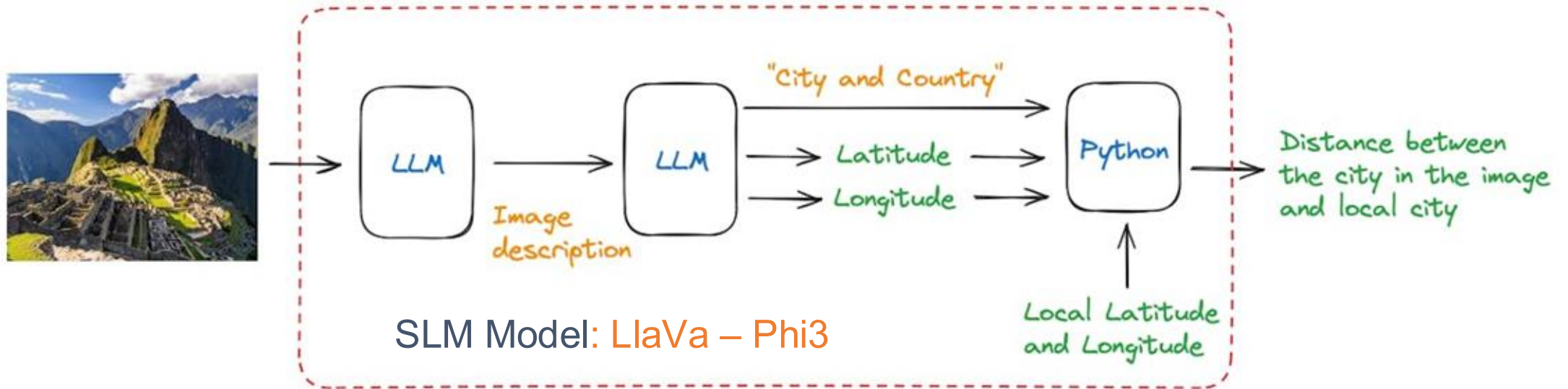
```
./build/bin/llama-server \  
-m      .../Qwen_Qwen3.5-0.8B-Q8_0.gguf \  
--mmproj .../mmproj-F16_0.8B.gguf \  
--host 0.0.0.0 --port 8081 \  
-c 4096 -t 4 -n 1024 \  
--jinja \  
--image-max-tokens 256 \  
--reasoning off --reasoning-budget 0
```

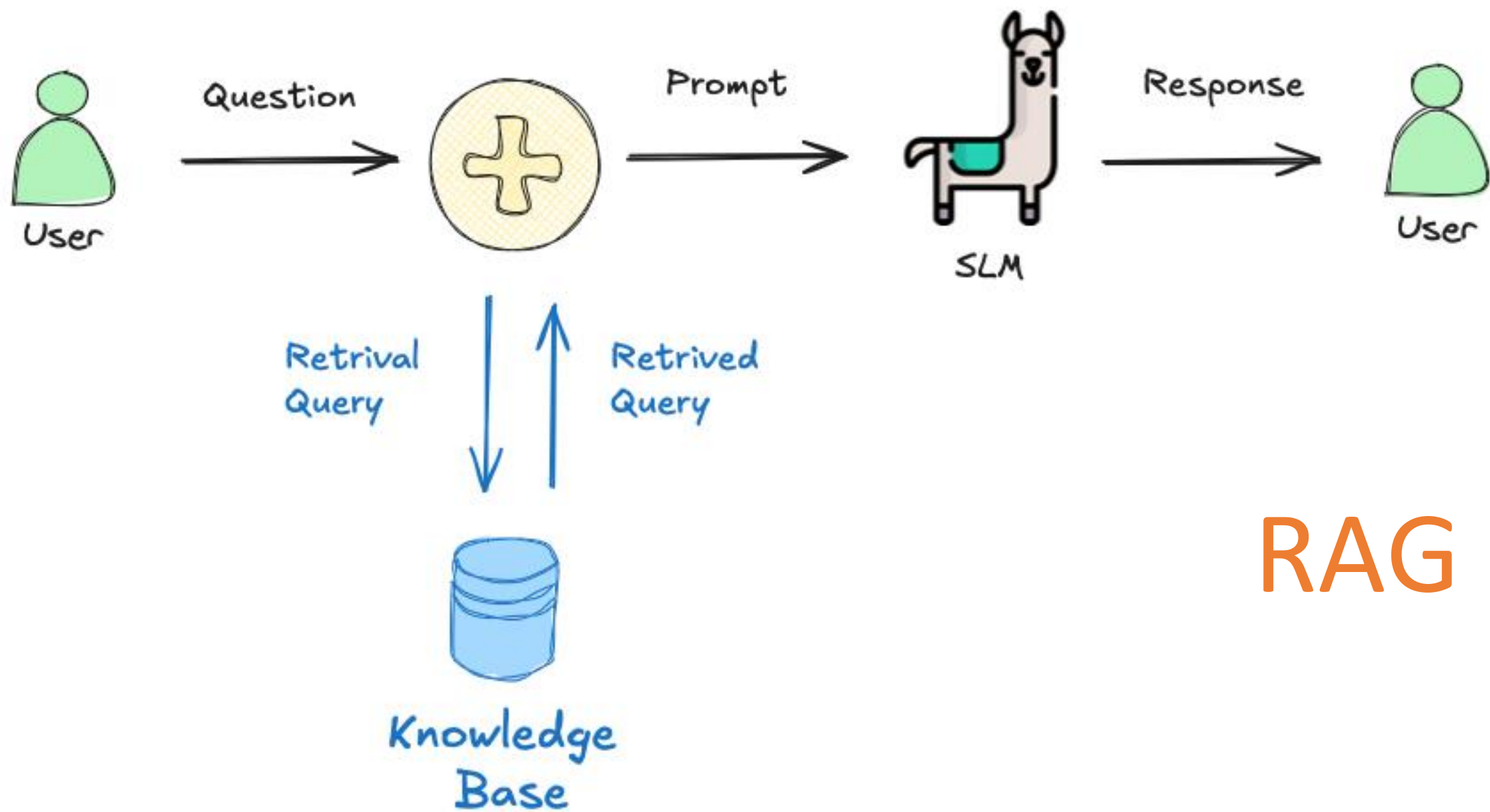


Agents



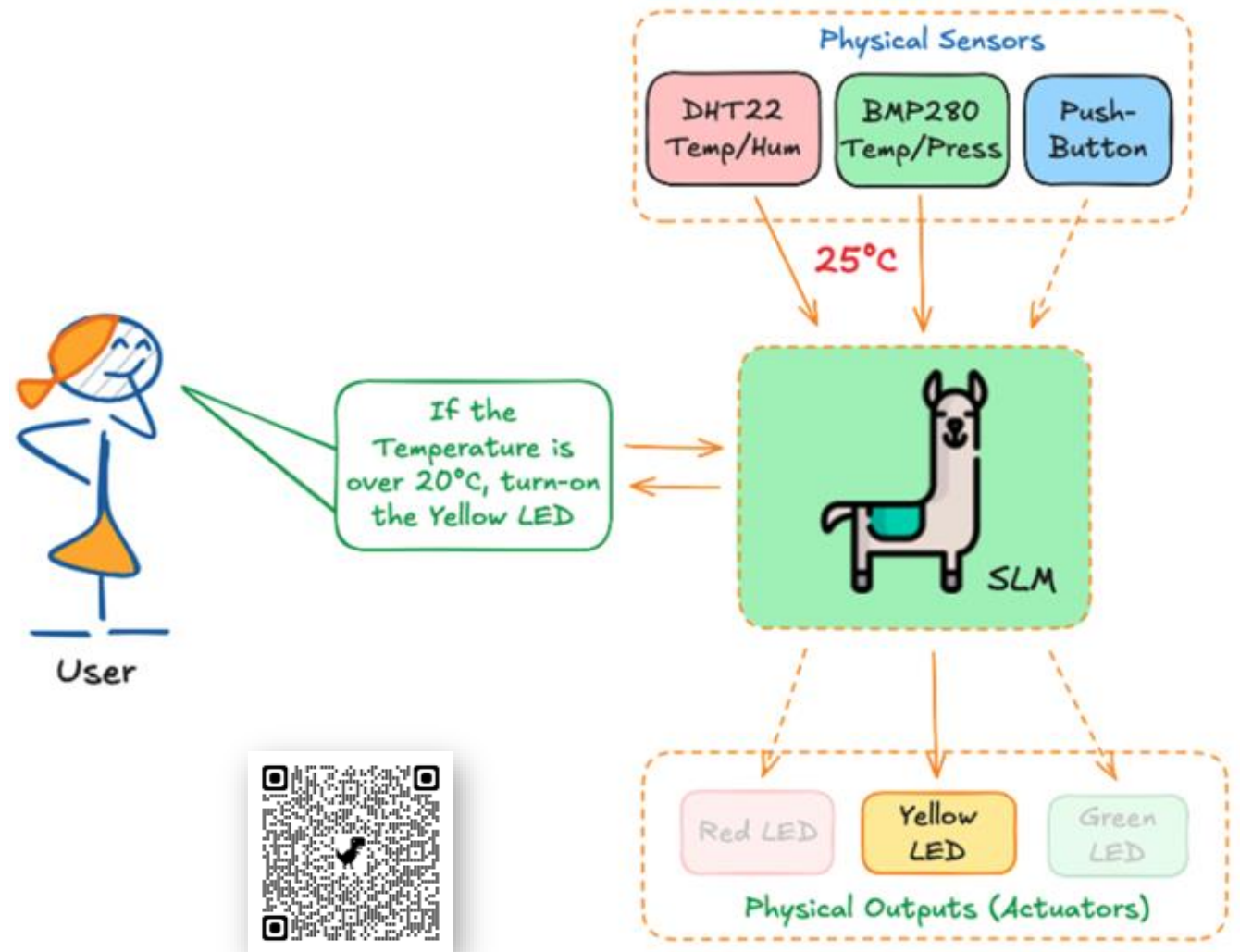
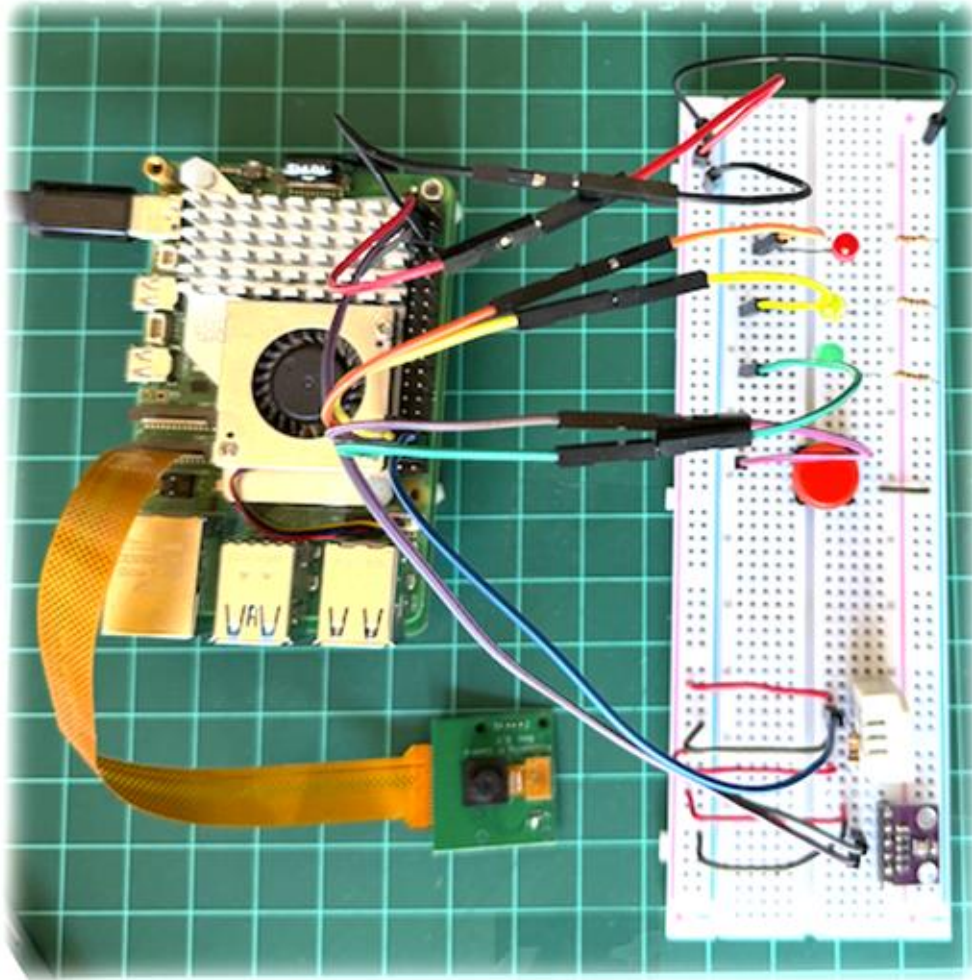
Function Calling





RAG

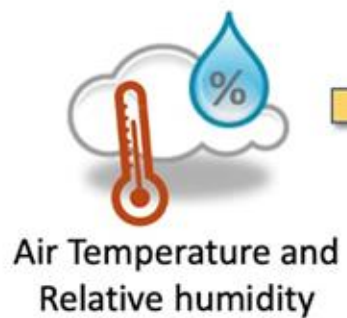
Physical Computing Meets Intelligence



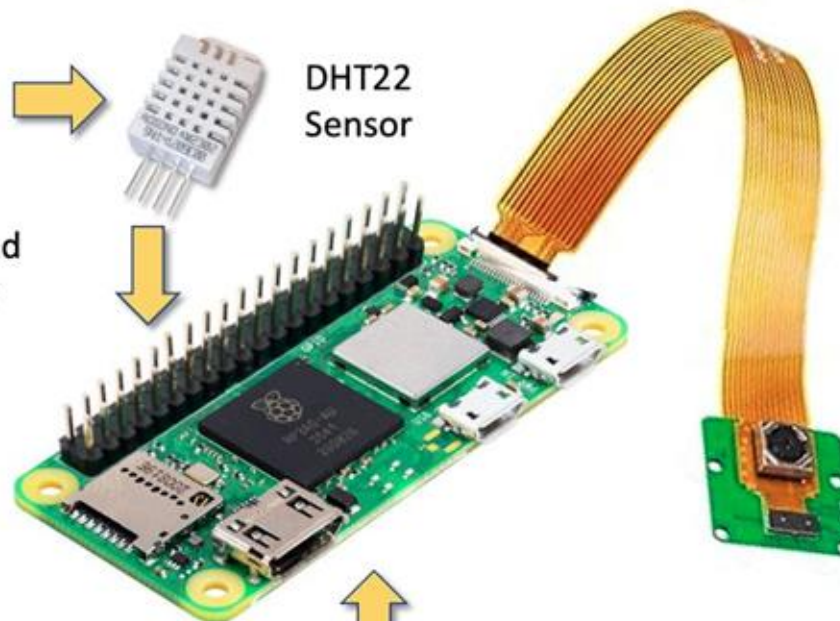
Real-World Applications



Bee Counting



DHT22
Sensor



Local
Database



sampleFreq → 10 s



José Anderson Reis
UNIFEI Master's Student

Guilherme Fernandes
UNIFEI Grad Student

Ant Detection



FURG



UTEC



UNRaf



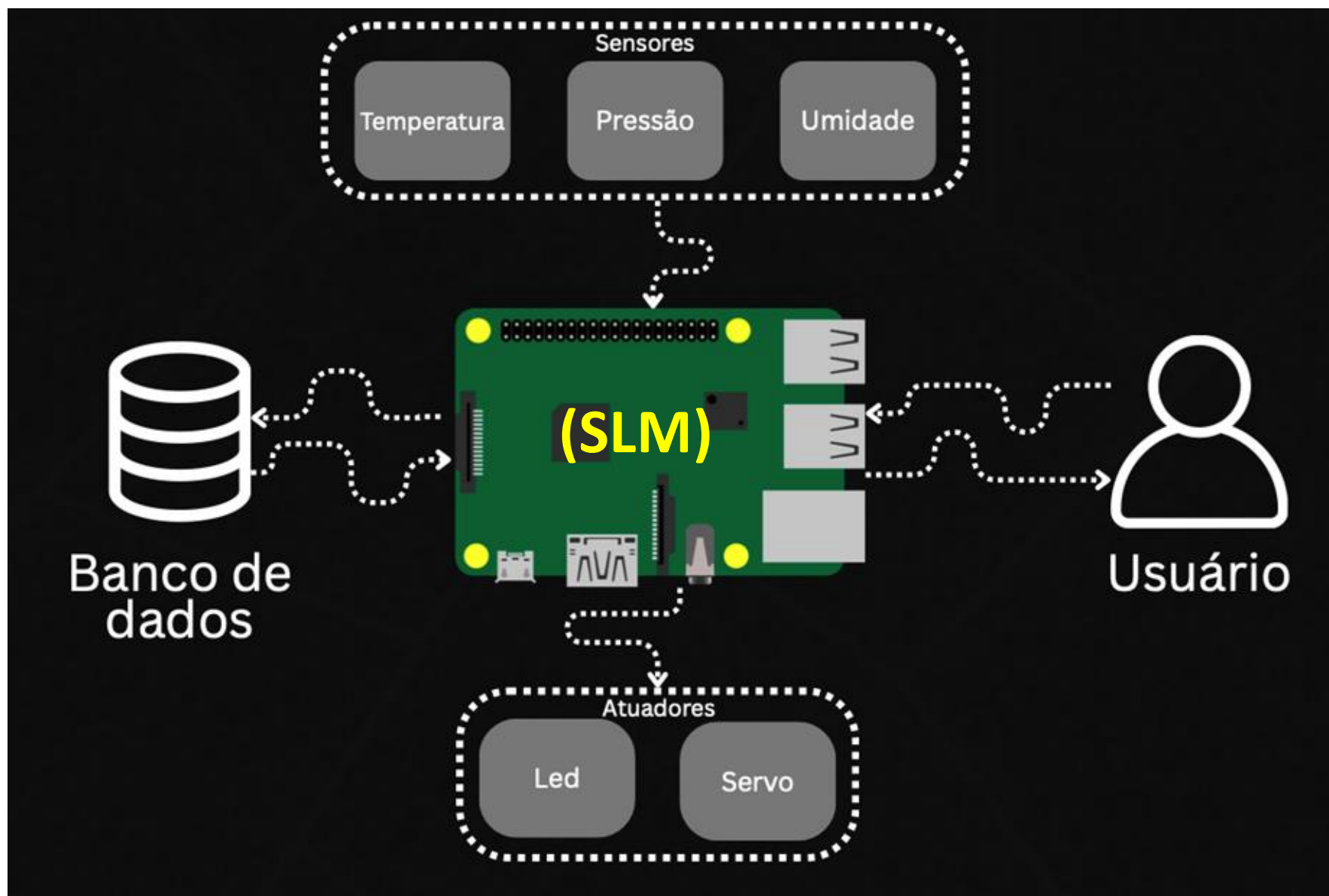
Juan Camilo Abedala
PRIA

Dança AI

Sistema
Embarcado de
Visão
Computacional
para Análise
Biomecânica e
Cadência Rítmica
em Dança de Salão



Felipe Santana Medeiros
João Luiz de Miranda Cilli
UNIFEI



João Pedro Fonseca
UNIFEI Grad Student (IESTI)

Matheus Souza
UNIFEI Grad Student (IMC)

Intelligent Colposcope Image Classification System



BRUNA CUSTÓDIO ALVES,
VIVIAN LEITE FRAGOSO
UNIFEI



Classe detectada : type1 (75.79%) | 147.2 ms
Enviando ao MedGemma...

Latência MedGemma : 63.0 s

1. ****Características esperadas da ZT1:**** A ZT1 é caracterizada por uma mudança de cor, geralmente de um tom rosado claro a um tom rosado mais escuro, com uma transição gradual entre as camadas epiteliais.

2. ****Visibilidade da JEC:**** A JEC deve ser visível, com uma transição clara entre o epitélio escamoso do ectocérvice e o epitélio colunar do endocérvice.

3. ****Risco associado:**** Risco baixo a moderado. A ZT1 é considerada um risco intermediário, com potencial de progressão para um câncer do colo do útero.

4. ****Conduta recomendada:**** Colposcopia com biópsia da ZT1 para avaliação histopatológica. Se a biópsia for negativa, acompanhamento com colposcopia em 6-12 meses. Se a biópsia for positiva, o tratamento (ex.: criocirurgia, laser, excisão cirúrgica) depende da extensão da lesão.



About the Course & Syllabus

Hands-on Learning

- **Software / Tools**

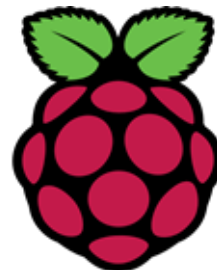
- Linux
- Python
- TensorFlow and PyTorch
- Google Colab or Jupyter Notebook
- Edge Impulse Studio
- Ultralytics and RoboFlow
- LangChain
- Ollama
- Llama.cpp

- **Hardware**

- Raspberry Pi Zero W2
- Arduino UNO Q
- Raspberry Pi 5
- Miscellaneous items such as LEDs, Sensors, etc.



Ollama



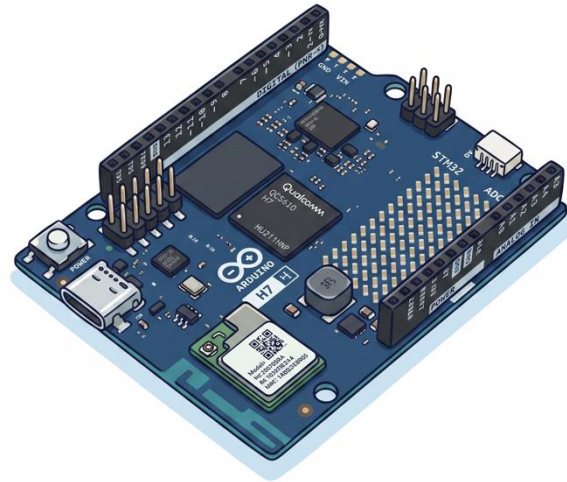
LangChain

Part 1: Fixed Function AI (Reactive)

Part 2: Generative AI (Proactive)



Raspberry Pi Zero W2

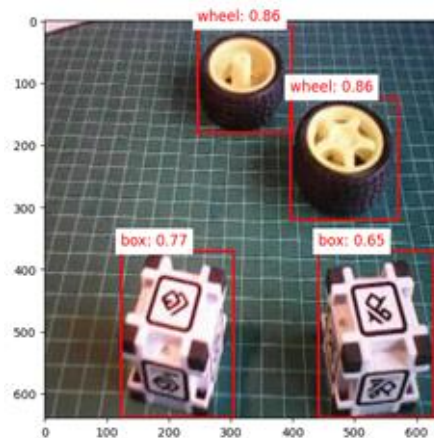
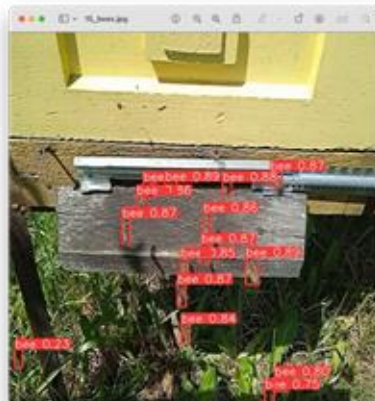


Arduino UNO Q



Raspberry Pi 5

Part 1: Fixed Function AI (Reactive)



Part 2: Generative AI (Proactive)

```
marcelo_roval — mroval@raspi-5: ~/Documents/Ollama/Rag/edgeai — ssh mroval@192.168.4.209 — 96x22

Your question: How to setup a Raspberry Pi?

Generating answer...

Question: How to setup a Raspberry Pi?
Retrieving documents...
Retrieved 4 document chunks
Generating answer...
Response latency: 137.56 seconds using model: llama3.2:3b

ANSWER:
=====
To set up a Raspberry Pi, download and install the Raspberry Pi Imager on your computer, select the operating system (e.g., Raspbian OS 32-bit or 64-bit), configure settings such as hostname, username, password, and SSH enablement, and write the image to an SD card. Insert the SD card in to the Raspberry Pi, connect power, and wait for the initial boot process to complete. You can then access the Raspberry Pi remotely using SSH or start a Jupyter Notebook server to control GPI Os.
=====

Your question: 
```

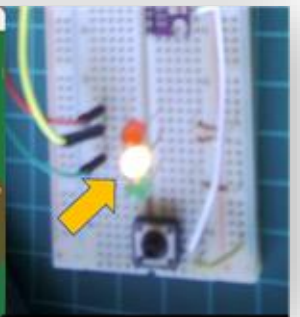
```
marcelo_roval — mroval@raspi-5: ~/Documents/Smart_bot/Basic — ssh mroval@192.168.4.209 — 87x17

Command: if the temperature is above 20°C, turn on yellow led.

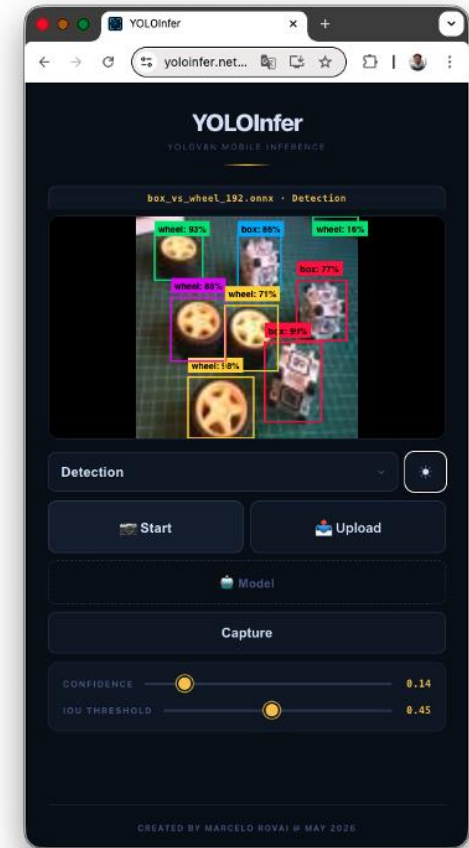
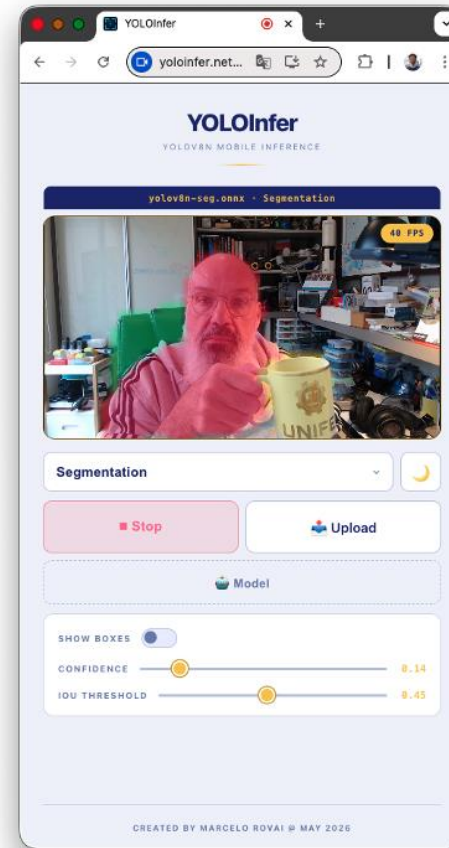
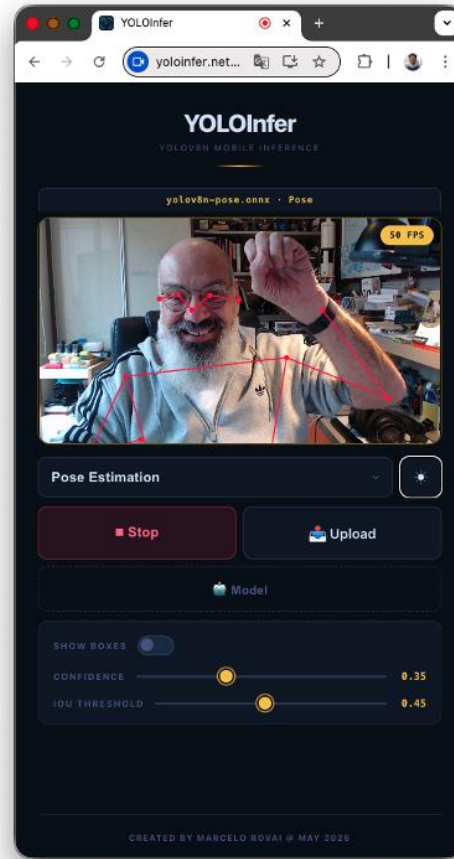
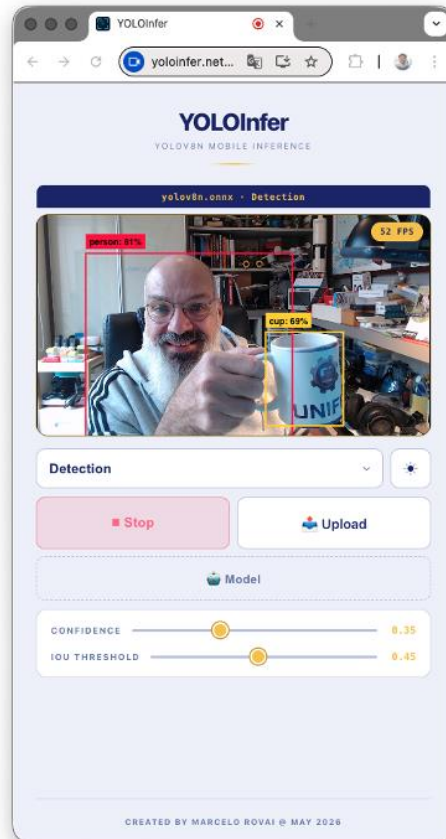
Time: 10:25:19
DHT22: 23.2°C, 42.4%
BMP280: 23.7°C, 905.2hPa
Button: not pressed
SLM Response: To turn on the yellow LED, as the temperature (23.7°C) is above 20°C, I will proceed with activating it.

Activate Yellow LED
LED Status: R=off, Y=ON, G=off

Command: 
```



YOLOInfer - Mobile YOLO Inference



Course Structure

The course is divided into two main parts (based on the textbook: [Edge AI Engineering by Marcelo Rovai](#)):

- **Part 1 (Weeks 1-7): Fixed Function AI** - Focus on image classification and object detection
- **Part 2 (Weeks 8-15): Generative AI** - Focus on Small Language Models (SLMs)

Key Learning Areas

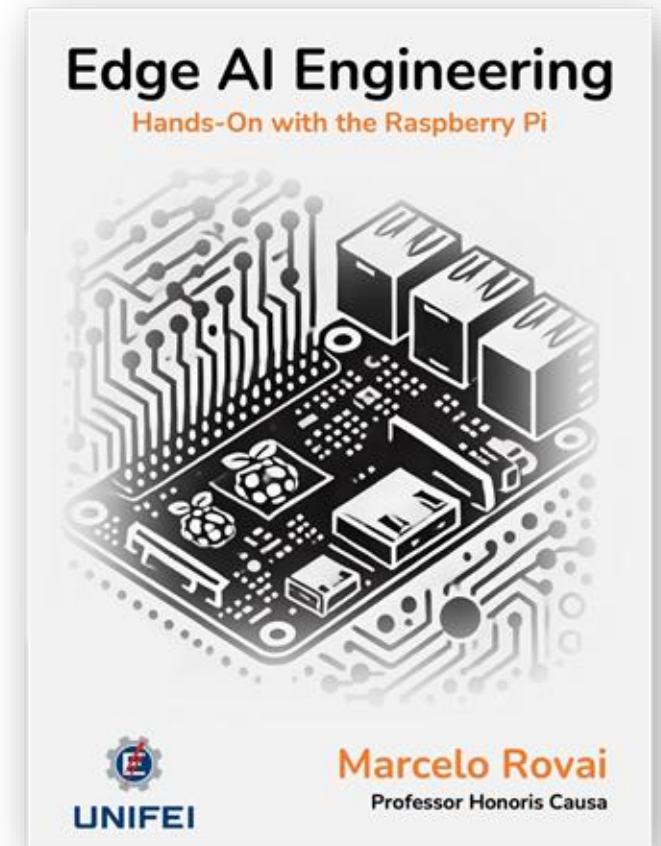
- Raspberry Pi and Arduino UNO Q setup, configuration, and optimization
- Computer vision and TensorFlow Lite
- Image classification and object detection implementation
- Small Language Model deployment and integration
- Retrieval-Augmented Generation (RAG) systems and Agents
- Physical computing integration with sensors and actuators

Assessment Structure

- **40%** - Weekly hands-on labs, quizzes, and surveys
- **20%** - Midterm project (Fixed Function AI system)
- **30%** - Final project (Generative AI application)
- **10%** - Participation and documentation



[EdgeAI Engineering](#)



Tópico: Week 1 | IESTI05_2025

https://moodle.unifei.edu.br/course/section.php?id=82817

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Página inicialPainelMeus cursos

Modo de edição

×

▼ IESTI05 - Summary

Avisos

▼ Pre-Course Activities

Suggested Review

Pre-Course Survey

▼ Week 1

▼ About the Course and Syll...

▼ Introduction to Edge AI

Lab 1a: Raspberry Pi Config...

Lab 1b: Development Enviro...

▼ Optional Readings

▼ Introduction to Edge AI

For this week's lab, please review the Book Chapter: [Setup](#).

For a quick review, please take a look at the video below, which covers the most essential Linux terminal commands, from navigating the file system to managing files, processes, and permissions directly from the command line. The video has an audio track in Portuguese. We recommend that you see it before starting work in the labs.



📄 Lab 1a: Raspberry Pi Configuration

Aberto: segunda-feira, 4 ago. 2025, 00:00 Vencimento: quarta-feira, 20 ago. 2025, 00:00

📄 Lab 1b: Development Environment Setup

Aberto: segunda-feira, 4 ago. 2025, 00:00 Vencimento: quarta-feira, 20 ago. 2025, 00:00

▼ Optional Readings

- [Cutting AI down to size, Science Magazine](#)
- [Generative AI at the Edge: Challenges and Opportunities, Vijay Janapa Reddi, ASM](#)

IES TI05_2025_S2_T01: Lab 1

+

← → ↺ 🏠

https://moodle.unifei.edu.br/mod/assign/view.php?id=216826

☆ 📁 👤 ⋮

Página inicial Pannel Meus cursos

   ⌵Modo de edição 

×

⋮

IES TI05 - Summary

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Week 1

About the Course and Syll...

Introduction to Edge AI

Lab 1a: Raspberry Pi Configur...

Lab 1b: Development Enviro...

Optional Readings

IES TI05_2025_S2_T01 - MACHINE LEARNING SYSTEMS ENGINEERING / Week 1 / Introduction to Edge AI

/ Lab 1a: Raspberry Pi Configuration

 **Lab 1a: Raspberry Pi Configuration**

Tarefa Configurações Envios Avaliação avançada Mais ⌵

Aberto: segunda-feira, 4 ago. 2025, 00:00

Vencimento: quarta-feira, 20 ago. 2025, 00:00

Objectives:

- Install Raspberry Pi OS using Raspberry Pi Imager
- Configure basic settings (hostname, SSH, WiFi)
- Learn essential Linux commands
- Manage files between your computer and Raspberry Pi

Instructions:

1. Download Raspberry Pi Imager on your computer
2. Configure OS settings (enable SSH, set hostname, WiFi credentials)
3. Boot your Raspberry Pi and confirm connectivity
4. Learn how to use SSH for remote access
5. Transfer files using SCP or FileZilla
6. Update your Raspberry Pi OS (`sudo apt update && sudo apt upgrade`)
7. Practice basic Linux commands (`ls`, `cd`, `mkdir`, `cp`, `mv`)

Deliverable: General comments and learning about the lab. Include a screenshot showing a successful SSH connection to your Raspberry Pi

Browser: mjrovai.github.io/TinyML4D/

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MARCELO ROVAI · PROFESSOR, UNIFEI

Edge AI & TinyML

A personal hub for my lectures, books, workshops, and tutorials — part of my work with the [AIEng4D academic network](#), bringing Edge AI & TinyML education across the Global South.

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Browser: mjrovai.github.io/TinyML4D/

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Marcelo Rovai — educator, professor, and co-chair of the AIEng4D network.
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- UNIFEI-IESTI01: TinyML - Machine Learning for Embedded Devices
- UNIFEI-IESTI05: EdgeAI - Edge Machine Learning Systems Engineering

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- Machine Learning Systems by Prof. Vijay Janapa Reddi (contributor)
- XIAO: Big Power, Small Board - Mastering Arduino and TinyML
- TinyML Made Easy: Hands-On with the Nicla Vision
- TinyML Made Easy: Hands-On with Seeed Studio Devices
- Edge AI Engineering: Hands-on with the Raspberry Pi

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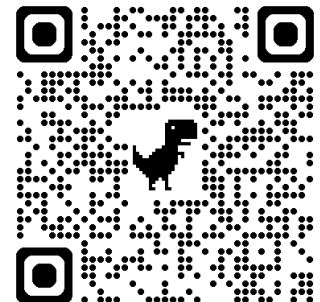
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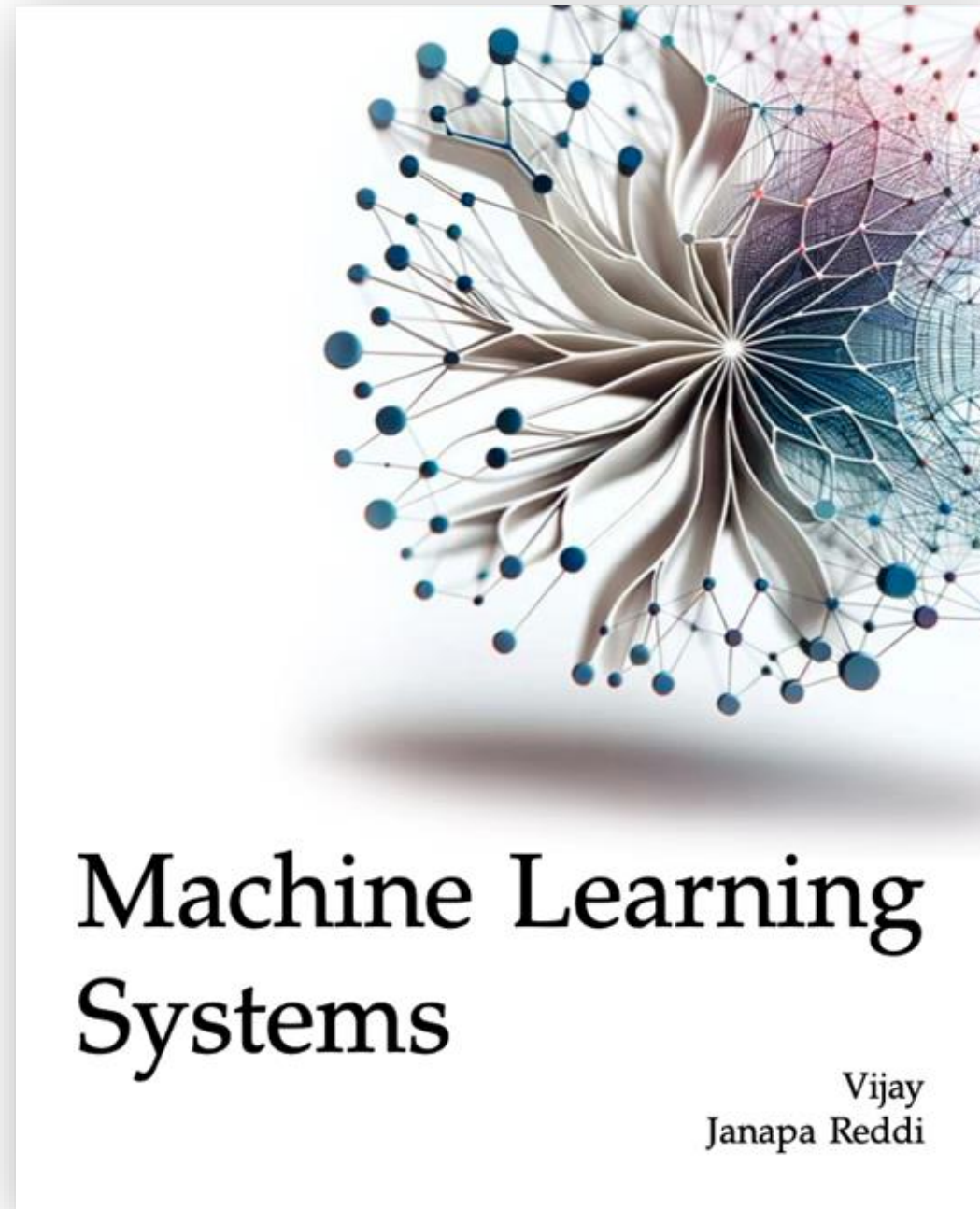


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Questions?



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